Innovation in Science Pursuit for Inspired Research (INSPIRE AWARDS)

Million Minds Augmenting National Aspiration and Knowledge (MANAK)

INSPIRE AWARDS – MANAK

Targeting one million original ideas/innovations rooted in science and societal applications.

To foster a culture of creative and innovative thinking among school children to address the societal need through Science and Technology.

Department of Science and Technology (DST), Government of India has been implementing the INSPIRE Award scheme since 2009 – 10. The present INSPIRE – MANAK is being modified suitably in collaboration with National Innovation Foundation (NIF), India to keep it in line with the Start-up India Initiative.

What is innovation?

Innovation can simply be described as a new idea, device or method, which is useful and scalable to appropriate levels. It should have certain distinctiveness over existing similar products/devices/methods, and which help improve output, efficiency, multi-functionality, reduce drudgery etc.

Why is innovation necessary?

Innovation is necessary not only to maintain competitive advantage but, for growth and development of institutions, industries and countries in a global world. Innovation contributes to improving the quality of life of people by improving products, processes and services. Thus it becomes imperative not only to remain ahead in cutting-edge research but also in innovations. Recognising the role of innovation in governance, the Government of India had declared 2010 – 20 as the "Decade of Innovation".

Why should we urge our children to be innovative?

Creativity among children is almost in-born. Every child is creative; the degree may vary, however, not the basic manifestation. Originality, creativity and innovative spirit among our children have to be promoted. This will be helpful when they become leaders of our society in future ensuring an imaginative and inclusive future for the country fuelled by innovation. Inclusive development of our society needs opportunities for children so as to ensure an aversion to inertia and a reverence for innovative ways of solving problems. Creative, impatient and imaginative children are one of the most precious assets of the nation. Inculcating the spirit of *samvedna* (empathy), *sarjansheelta* (creativity) and *sahyog* (cooperation) will make them worthy citizens of not just India but the world.

What is INSPIRE AWARDS – MANAK?

INSPIRE is one of the flagship programs conceptualised and developed by the DST. The basic objective of INSPIRE is to attract talent, have them exposed to science at an early age (10 – 15 years or Standard 6-10) and help towards building a critical human resource pool for strengthening and expanding the science and technology system and increase the research and development base. Students are invited from all government or private schools throughout the country to send their original and creative technological ideas/innovations for the same.

What kind of submissions are invited under INSPIRE AWARDS – MANAK?

Original and creative technological ideas/innovations of students that solve any daily problem, be it household or for farmers, porters, labourers, society or the like. Children may be urged to look around for common problems and come up with solutions on their own.

How will the scheme be operationalised?

- Awareness and capacity building of District and State level functionaries across the country through regional workshops, audio-visual tools and literature.
- Organising internal competitions in schools and nominations of two to three best original ideas, in any Indian language, by respective the Principal/Headmaster online through E-MIAS (E-Management of INSPIRE Award Scheme) portal on web-link http://www.inspireawards-dst.gov.in.
- The schools may also submit students ideas/innovations through NIF website on web-link http://nif.org.in/inspire/2017/submit-inspire-idea.
- Only the schools located in northeast region (Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura) and Jammu & Kashmir may also send ideas/innovations by speed post in prescribed format to National Innovation Foundation – India, as "INSPIRE-MANAK Submissions".
- Shortlisting of top 1,00,000 (one lakh) ideas, with potential to address societal needs through Science & Technology, by National Innovation Foundation (NIF). Disbursement of INSPIRE Award of INR 10,000 into bank accounts of students through Direct Benefit Transfer (DBT) scheme.
- Organisation of District Level Exhibition and Project Competition (DLEPC) by District/State authorities and shortlisting of 10,000 best ideas/innovations for State Level Exhibition and Project Competitions (SLEPC).
- Organisation of State Level Exhibition & Project Competition (SLEPC) for further shortlisting of top 1,000 ideas/innovations. Disbursement of INR 20,000 to student of each idea/innovation through DBT for development of improved prototypes for the National level event. Selection of ideas/innovations will be based on novelty, social applicability, user and eco-friendliness and advantage over the existing technologies.
- Mentoring support to students at State level by NIF, in coordination with reputed academic and technology institutions of the country for development of prototypes.

- Showcasing 1,000 best ideas/innovations at National Level Exhibition & Project Competition (NLEPC) and shortlisting of top 60 prototypes/innovations for national awards and future direction.
- Consideration of top 60 ideas/innovations by NIF for product/process development and their linkage with other schemes of NIF/DST and display of top innovations at the annual Festival of Innovation (FOIN) at Rashtrapati Bhavan.

Who can participate?

Students of any age group of 10 - 15 years and studying in standard 6-10 will be eligible for participation.

Can students of all educational boards (national and state) participate in the competition?

Yes, students irrespective of their educational board (national and state) can participate in the INSPIRE AWARDS – MANAK scheme.

In which can students submit their entries?

Students can submit their entries to their school in any language, recognised by the Constitution of India.

What is the contest period?

Online nomination for the year 2017-18 is open till 15 August 2017.

What kinds of submissions are not encouraged?

Examples include but are not limited to be following:
Common ideas/concepts (from textbooks/others) like hydel power projects, rain water harvesting, water level indicator, vermi compost/vermin wash, letter box alarm, other such alarm, using turbines to generate energy, electricity generation through waste batteries/dung/transport/wave or earthquake alarms, etc.
Children should be encouraged to come up with creative ideas on their own. Parents or teachers may propose a problem to the child or help in converting his/her idea into a prototype/model but not suggest the idea themselves.

The teachers may also look upon for INSPIRE (http://www.inspireawards-dst.gov.in/UserP/inspire-downloads.aspx) to see if the projects being developed by their wards have been not recognised already. This will help ensure originality and maintain the standard of the competition.

What is the process of idea/innovation submission?

The principal/headmaster of the school will call for submission of ideas in the form of synopsis from the students of the designated classes and age groups. Students may thus

submit their ideas to the school. For this purpose, an Idea Competition can also be conducted in Schools.

What is an Idea Competition?

This competition can be organized by assembling students and enabling them to think of an idea/innovation such as

- A machine or a gadget, which is not available but the student's desires to have such a machine or a gadget,
- Improvement required in any existing/available machine or gadget, so that it becomes multifunctional, improves efficiency or output, reduces drudgery etc.An idea that can solve a local technological problem, which a student may be seeing everyday around him/her.

After the students have thought of ideas, they may note down their ideas on a piece of paper and submit it to the Principal/Headmaster/Teacher. The submitted ideas may be reviewed at the school level and shortlisted ones submitted for INSPIRE-MANAK Awards as per the process described above.

Is there any limit to the number of ideas submitted by the students?
No, the students are encouraged to submit as many ideas as they wish.

How will the Principal/Headmaster submit the nominations?

All nominations can be submitted online by the Headmaster/Principal through E-MIAS (E-Management of INSPIRE Award Scheme) portal on DST (http://www.inspireawards-dst.gov.in). New schools may also register themselves on this portal.

The Principal/Headmaster may also submit selected ideas/innovations through http://nif.org.in/inspire/2017/submit-inspire-idea. Only the schools located in northeast region (Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura) and Jammu & Kashmir may also send ideas/innovations by post to National Innovation Foundation – India, Grambharti, Amrapur, Gandhinagar-Mahudi Road, Gandhinagar, Gujarat Pin: 382650 in an envelope superscribed as, "INSPIRE-MANAK Submissions".

How will the screening and shortlisting be executed?

The selection criteria will be based on innovation, creativity and novelty of the ideas/innovations that a student proposes to undertake, as given in the submitted synopsis. DST/NIF will ensure that there is due representation of girl students and SC/STs.

What should the principals and jury members bear in mind while selecting ideas/innovations?

The degree of novelty of the idea, its social applicability and end use, environmental implications, potential impact, scope for diffusion through commercial and/or non-commercial channels, relevance to existing government schemes etc. may be considered.

Are there any separate awards for individuals or groups?

No, there are no separate awards for the same.

Are any photos, videos, sketches of the idea/innovation required?

Yes, if available. These would help our experts to properly understand and evaluate the submission.

When will the awards be announced and given?

- Inviting nominations as per revised format till 15 August 2017
- Processing of the nominations and disbursements of funds -------
- Organisations of D/SLEPC-------
- Organisations of NLEPC 1ST Week of December
- Display of top 60 ideas/innovations at the festival of innovation in March 2018

How will the award money be disbursed?

Through Direct Benefit Transfer (DBT) scheme in students' bank accounts.

What are the various activities that will be undertaken in the regional and innovation workshops?

In the regional workshops, organised prior to the submission of entries for the competition, the participants (comprising teachers, principals, district and state-level functionaries) will be informed about the revised scheme and sensitised about the need for inculcating innovative and original thinking in our students.

In the innovation workshops at various R&D and engineering institution, organised before the national level event, the focus will be on teaching the children basics about prototype development and improving the quality of products displayed at national-level exhibition. The children will also learn in the process through interactions with engineering and design students and faculty of institutions.